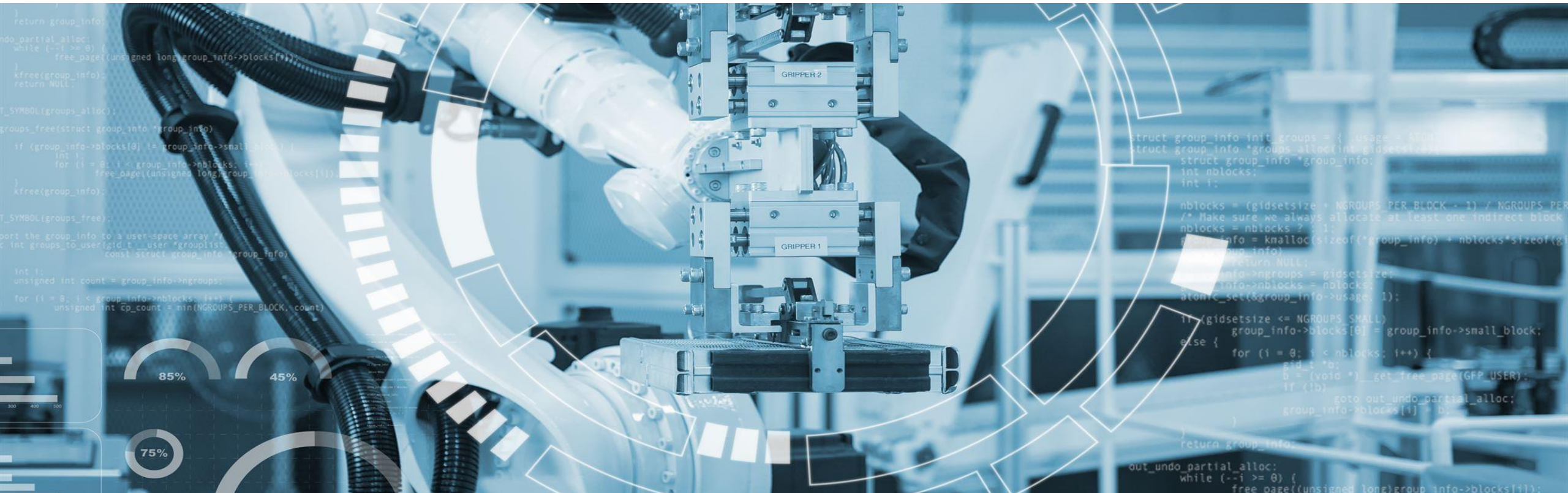


Project3 Proposal of CS_527: Data Science

HOOD COLLEGE



Predictive Maintenance of Industrial Equipment Using Machine Learning

Leveraging AI to forecast and prevent equipment failures

Date : April 29, 2026

By: 1. Nay Lin Aung (CS)
2. Han Soe (MIS)



Project Description:

This project builds and compares machine-learning models to predict imminent machine failures in an industrial manufacturing process using the AI4I 2020 Predictive Maintenance dataset.

Reference: S. Matzka, "Explainable Artificial Intelligence for Predictive Maintenance Applications," 2020 Third International Conference on Artificial Intelligence for Industries (AI4I), 2020, pp. 69-74, doi: 10.1109/AI4I49448.2020.00023.

Problem Statement and Project Motivation:



Challenges of Unplanned Failures

Unplanned equipment failures cause costly downtime, safety risks, and production delays in manufacturing environments.

Limitations of Traditional Maintenance

Reactive and preventive maintenance often fail to optimize machine uptime due to delayed or unnecessary interventions.

Predictive Maintenance Solution

Using machine learning on sensor data, predictive maintenance anticipates failures for proactive intervention and efficiency.

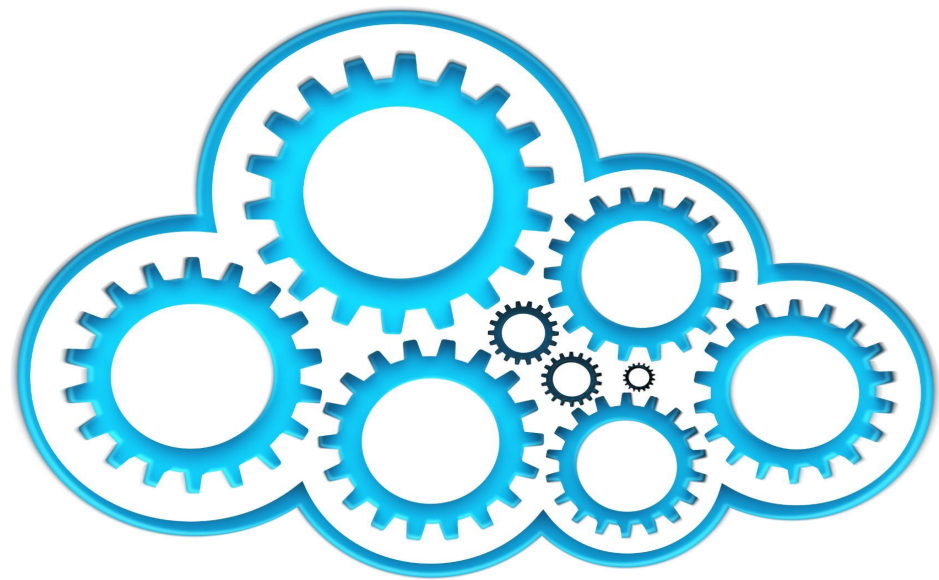
Project Objective and Relevance

The project aims to reduce risks and improve efficiency by predicting machine failures using non-linear models and classification metrics.

Data Source:

<https://www.kaggle.com/datasets/stephanmatzka/predictive-maintenance-dataset-ai4i-2020/data>

Applied Methodology and Modeling Approach



Data Preprocessing and Feature Engineering

Scaling numerical features and encoding categorical variables prepare data for machine learning algorithms. Interaction terms enhance model insight.

Model Selection and Comparison

Multiple supervised models including logistic regression, Random Forest, and Gradient Boosting will be compared for performance and complexity.

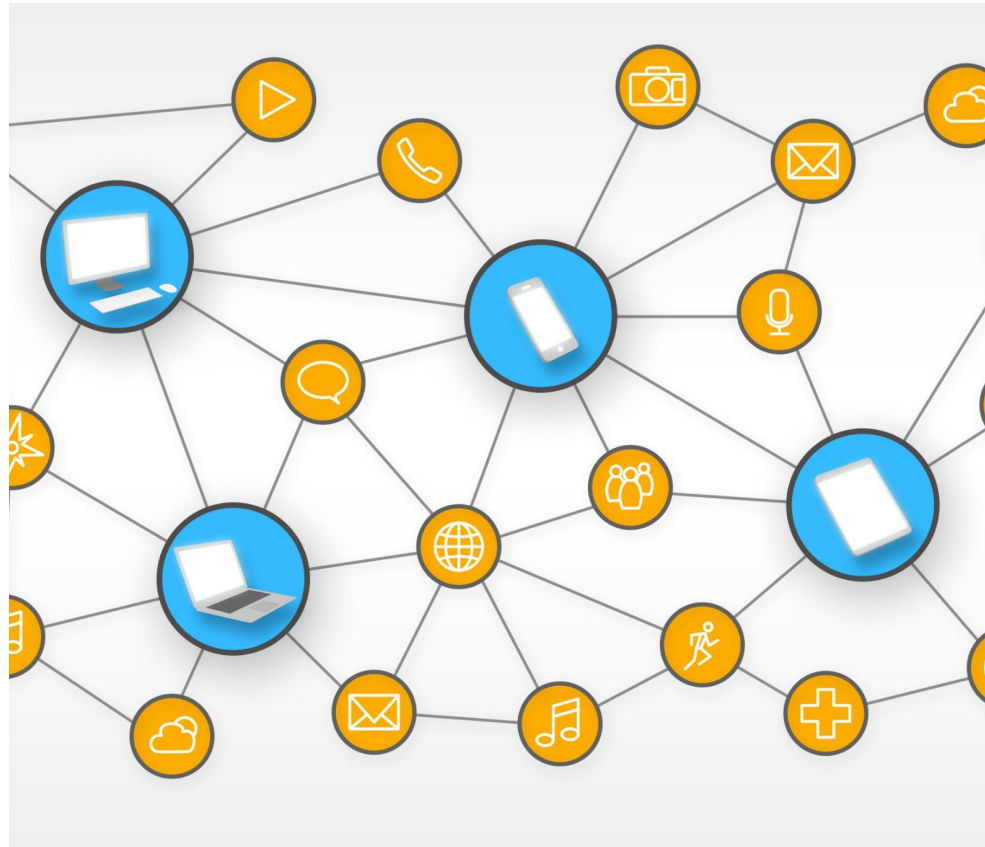
Evaluation Using Classification Metrics

Model evaluation focuses on precision, recall, F1-score, and ROC-AUC, emphasizing recall to reduce costly missed failure detections.

Threshold Tuning and Cross-Validation

Threshold tuning and cross-validation optimize model trade-offs to select the best predictive maintenance model.

Tools Used



Data Science Tools

The project primary uses Python with pandas, scikit-learn, XGBoost, and LightGBM for data manipulation and advanced modeling.

Visualization and Development

Matplotlib and seaborn generate clear plots, while Jupyter Notebook supports documented step-by-step analysis.

Dataset Description and Industrial Context:



Dataset Composition and Size

The dataset contains 10,000 observations with 14 variables representing machine operating conditions and failure labels.

Relevant Industrial Sensors

Features include air temperature, process temperature, rotational speed, torque, and tool wear from real manufacturing sensors.

Predictive Maintenance Challenge

Dataset models predictive maintenance with imbalanced failure classes and multiple correlated sensor variables.

Clean Dataset Advantage

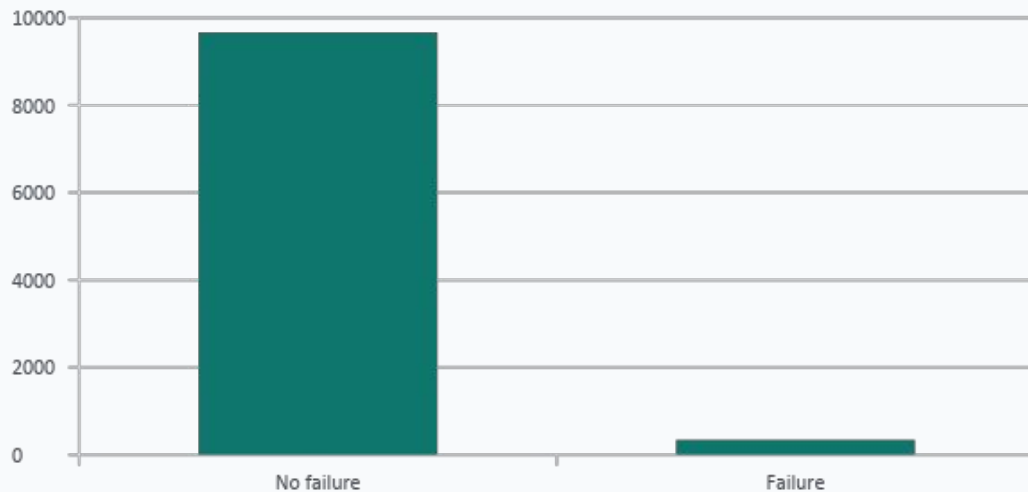
The dataset contains no missing values, enabling focus on modeling and analysis without data cleaning.

Observation of dataset

A rare-event decision problem, not an accuracy contest

Why this matters

- Unplanned failures create downtime, scrap, safety risk, and reactive maintenance cost.
- Only 339 of 10,000 rows are failures; standard accuracy is misleading.
- Useful alerts must balance missed failures against false alarms.
- Opportunity: earlier intervention using sensor-derived risk signals.



Class imbalance in source data

Sources: CSV target distribution; notebook evaluation framing.

No failure: 9,661 | Failure: 339

3.39%

FAILURE RATE

Rare positive class

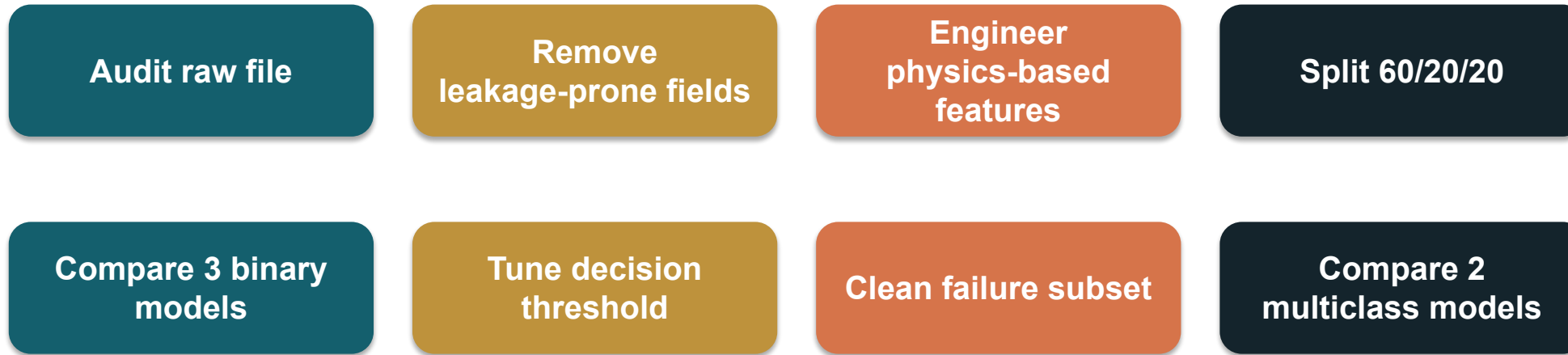
PR-AUC

SUCCESS LENS

Precision, recall, F1

Modeling Workflow

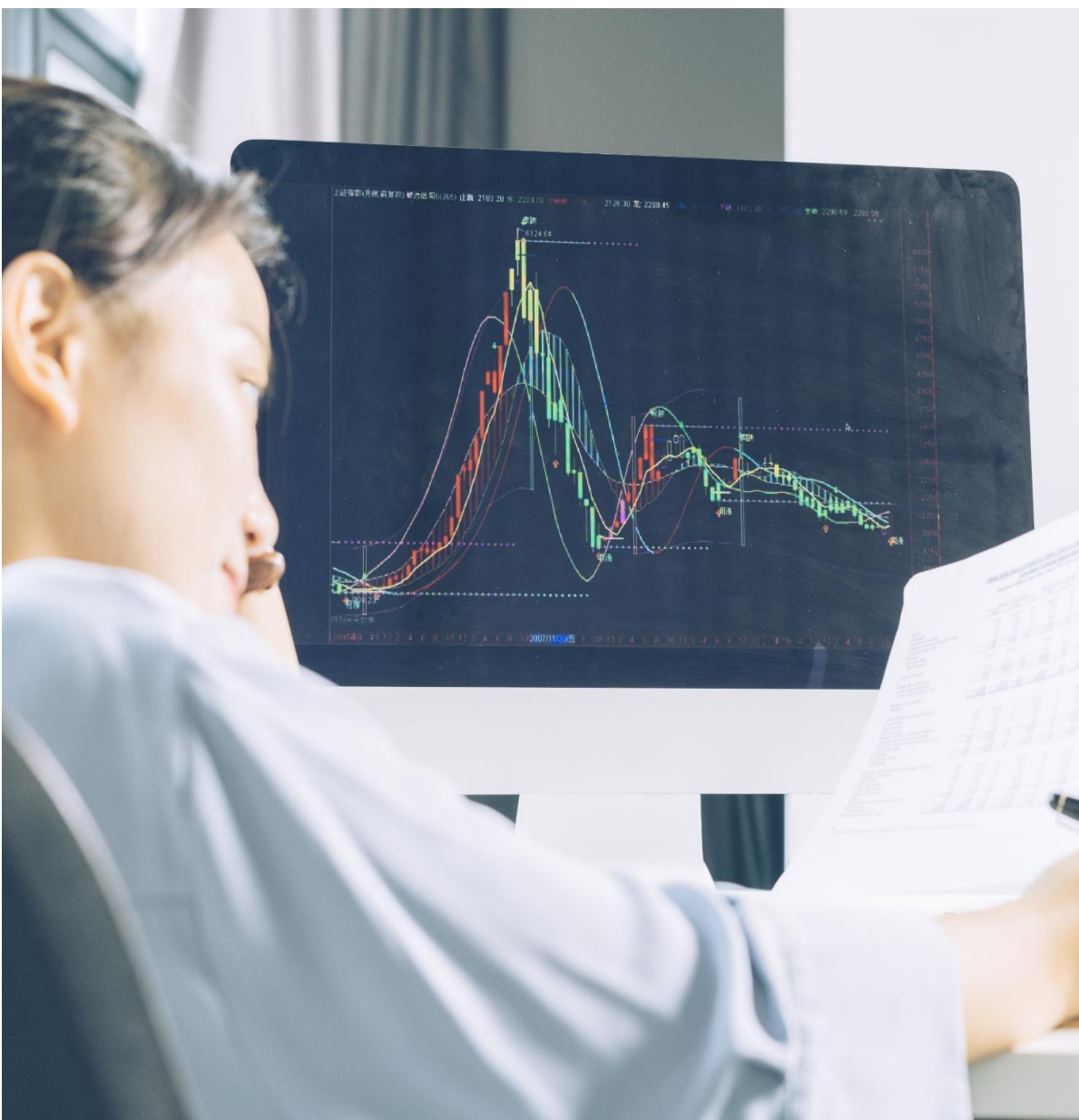
End-to-end analysis path used in the notebook



Binary comparison: Logistic Regression, Random Forest, and XGBoost.

Final binary selection rule: highest validation PR-AUC, with simplicity preferred if results are within 0.01.

Multiclass comparison: Multinomial Logistic Regression versus Multiclass XGBoost on clean single-cause failures.



Model Evaluation and Explainability

Comparative Model Evaluation

Models are compared using evaluation metrics and visual tools like confusion matrices and ROC curves to assess performance.

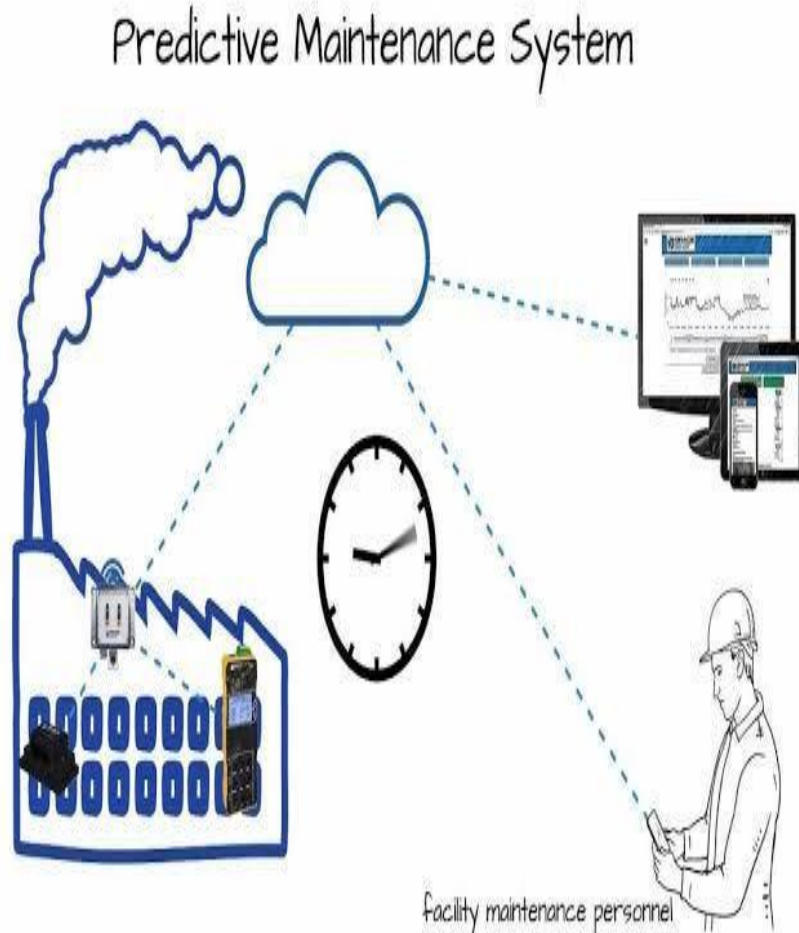
Explainability Techniques

Explainability methods clarify complex models by showing feature importance and prediction contributions for transparency.

Actionable Insights

Explainability helps identify key failure drivers, enabling engineers to trust and act on model recommendations.

SHAP Analysis

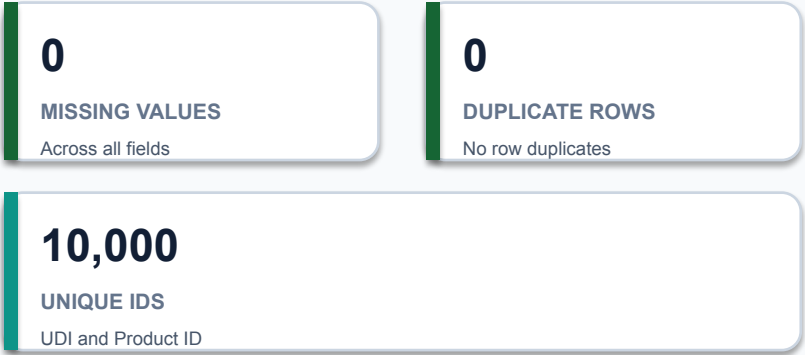


- SHAP analysis identified Torque, Tool Wear, and Temperature Difference as the top failure drivers.
- Engineered features improved minority failure detection significantly.

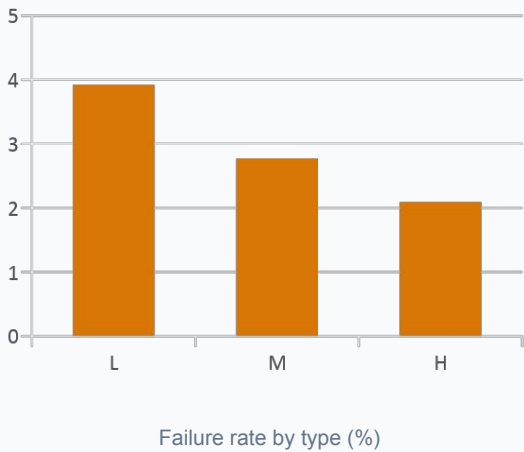
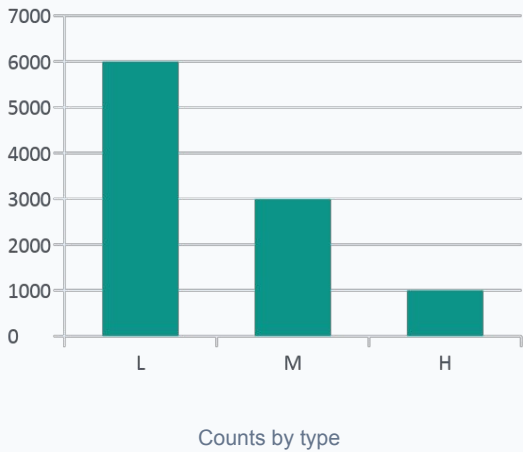
Key Finding 1: The Data Is Clean but Imbalanced

Quality is high; modeling discipline still matters

Data quality facts



Product mix and failure rate



Key Finding 2: Engineered Features Improve Signal

The strongest features translate raw sensors into operating mechanics

Physics-informed feature engineering

Feature	What it captures
Temp diff [K]	Heat dissipation condition
Power [W]	Torque × rotational speed
Wear × Torque	Mechanical stress from tool load

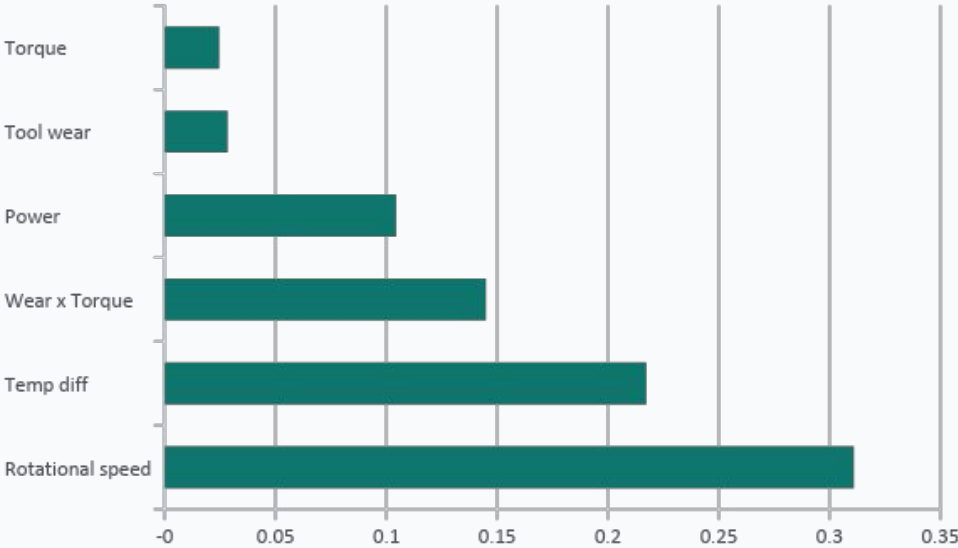
0.8468

BEST VALIDATION PR-AUC
Engineered Random Forest

0.8739

TEST PR-AUC
Same selected model

Top permutation-importance drivers



Key Finding 3: Strong Performance, Not Perfect

Final binary model catches most failures with few false alarms

Final model and threshold

Random Forest

MODEL

Engineered features

0.40

THRESHOLD

Selected on validation

0.855 / 0.889 / 0.824

F1 / PRECISION / RECALL

Tuned test results

Test confusion matrix

	Predicted no	Predicted yes
Actual no	1,925	7
Actual yes	12	56

Primary operating risk

12 missed failures remain in the test split.

Challenges Encounter



- Highly imbalanced dataset with very few failure observations.
- Presence of identifier columns that could create data leakage.
- Difficulty predicting minority failure modes accurately.
- Lack of real timestamped production data for live maintenance simulation.

Challenges Solved & Unsolved

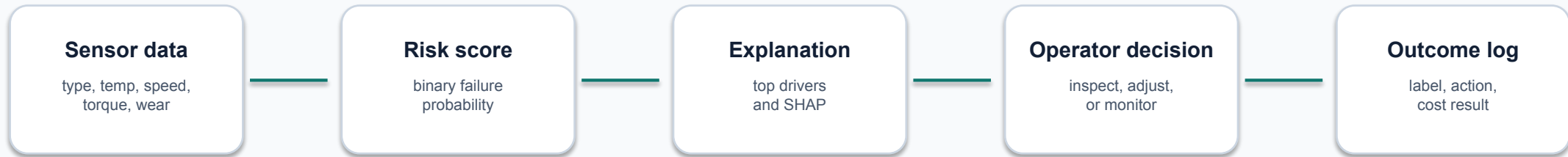


- **Solved**: Used stratified splitting and F1/ROC-AUC instead of accuracy for fair evaluation.
- **Solved**: Removed leakage variables and engineered physically meaningful features.
- **Partially Solved**: Ensemble boosting improved minority failure prediction but rare events remain difficult.
- **Not Yet Solved**: Real-time deployment and concept drift monitoring require live industrial IoT data.

Recommended Usage

Run a controlled, human-in-the-loop pilot before production

Recommended operating model



Implementation guardrails

- Use alerts for decision support, not automatic shutdown.
- Track false negatives, false positives, and maintenance actions.
- Use explanations to improve operator trust and reviewability.
- Defer production until live timestamped data confirms value.

Executive suggestion

Approve pilot funding and data access; do not approve autonomous production deployment yet.

Conclusion



- The project successfully demonstrates that machine learning can forecast machine failure proactively.
- LightGBM and XGBoost were the strongest models for industrial predictive maintenance.
- This framework offers a practical AI foundation for reducing downtime and improving manufacturing reliability.
- Future work will focus on real-time sensor integration and cost-sensitive maintenance optimization.

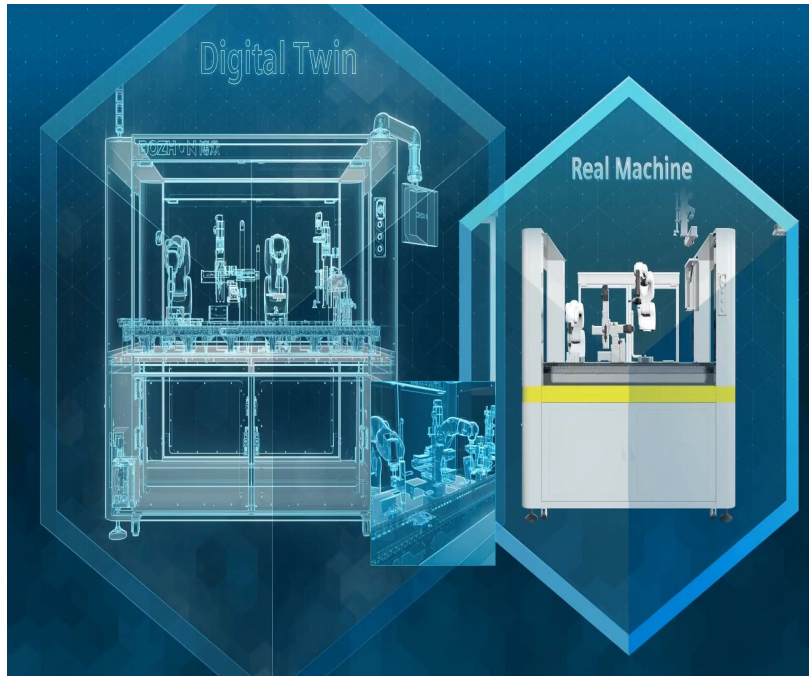
Conclusion: Business Impacts



Business View

- This project demonstrates that machine learning can move maintenance strategy from **reactive repair** to **proactive prediction**.
- Early failure detection helps manufacturers:
 - reduce unexpected downtime,
 - lower emergency maintenance costs,
 - improve equipment lifespan,
 - and increase production reliability.
 - extend customer loyalty

Conclusion: Digital Twin



Digital Twin / Industry 4.0 View

- The predictive model can serve as the **analytics brain of a Digital Twin**, where each physical machine has a live virtual replica.
- By continuously feeding sensor data into the trained model, the digital twin can:
 - simulate machine health in real time,
 - estimate failure probability,
 - trigger predictive alerts,
 - and support autonomous maintenance planning.
- This creates a foundation for **smart factories**, where AI-driven monitoring improves operational efficiency and supports data-based industrial transformation.

Thank you for your attention!